THOMAS HOWARD

CTO & HEAD OF APPLIED AI | AGENTIC SYSTEMS | HEALTHCARE VOICE AI | AI EVALUATION & SAFETY

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PROFESSIONAL SUMMARY

Hands-on Applied AI leader who builds **production LLM agents and the controls that make them safe to ship**. Sole Applied AI engineer at a 10-person healthcare company, also leading its small engineering team: conceived, built, and shipped its AI product line from scratch, from **Sally**, a patient-facing voice agent now live in production, to **Symphony**, the governed system of 10 specialized engineering agents used to build its products. Previously led a **12-person engineering organization** for seven years. Builds with agents: sets the architecture, writes the task contracts, and reviews every change, in Python, TypeScript, and Go. Founded and built a production SaaS product end to end.

// EXPERIENCE

Chief Technology Officer & Head of Applied AI | Snap eHealth

Jun 2026 – Present

Promoted to CTO of this 10-person healthcare SaaS company, which makes referral and Annual Wellness Visit (AWV) software for primary care clinics, after four years building its AI platform; own **Applied AI strategy, product architecture, and engineering delivery**.

- Lead **Sally, an LLM-powered healthcare voice agent** live in production at pilot scale, running patient outreach calls for primary care clinics: real-time browser and phone conversation with interruption handling, tool execution, background analysis, and structured visit-brief extraction.
- Designed the boundary between model behavior and **deterministic consent, identity verification, authorization, and tool access** in patient-facing workflows.
- Built **scenario-based evaluation harnesses** covering conversational behavior, structured outputs, safety rules, and failure handling as part of Sally's production hardening.
- Architected **Symphony**, coordinating **10 specialized engineering agents** across implementation, testing, review, debugging, and remediation with isolated execution, signed leases, tool permissions, cost accounting, and audit trails; humans approve every merge and deploy.
- Built **four applications in four months** on my own with Symphony, across enterprise healthcare, workflow automation, and patient engagement, all live in production; Sally alone serves **~1,200 users across 4 primary care clinics**, with no production incidents to date.
- Set the Applied AI roadmap with the CEO and Product; hired and led a team of three engineers and now lead a senior engineer, training him in AI-assisted development, while staying hands-on in **Python, TypeScript, APIs, and healthcare integrations**.

Head of AI Platform | Snap eHealth

Oct 2022 – Jun 2026

- Led LLM reliability on **Amazon Bedrock**, building **evaluation workflows, goldset reference datasets, and guardrails**.
- Designed an **AI engineering control plane**: specialized monitoring, security, and SRE agents producing structured evidence, with deterministic triage, bounded remediation, protected verification gates, and human escalation.
- Built an **AI test-intelligence system** that decomposed product stories, generated test requirements and API tests, analyzed defects, and targeted retests.
- Architected an **explainable physician-referral ranking engine** in Python and scikit-learn, combining clinical quality, outcomes, eligibility, availability, and proximity with deterministic scoring, uncertainty handling, and abstention.
- Developing supervised models to **predict 30-day referral completion**, with leakage prevention and evaluation on precision, recall, ROC-AUC, and calibration.

Founder & Software Engineer | BusyNow | busynow.app Oct 2025 – Present

- Built and operate a production consumer SaaS end to end: **React/TypeScript and Node.js/Express** with REST APIs, PostgreSQL, and Redis.
- Use **Codex and supervised agent workflows** for implementation, test generation, debugging, and review, with CI verification and human-controlled releases.
- Designed **idempotency, retries, data-correction paths, rate limits, and provider budgets** so the product degrades gracefully under failure and keeps operating costs in check.
- Run the AWS runtime on **ECS Fargate**, migrated from Kubernetes to fit the product's scale, with VPC, IAM, CloudFront, and S3 defined in Terraform, and deployments with health checks and rollback.

Director of Engineering | FBS Jan 2015 – Apr 2022

- Built and led a **12-person engineering organization** at this startup, through an engineering manager, owning hiring, technical direction, mentoring, performance management, and leadership development.
- Reduced regression cycle time by **approximately 90%** through an automation-first transformation, earlier validation, and repeatable release controls.
- Directed CI/CD, developer tooling, AWS architecture (EC2, RDS, ECS), and production observability across multiple product lines with product and engineering leadership.

EARLIER EXPERIENCE

Software Engineer | **Google (Contract)** Jun 2012 – Dec 2014

Software Engineer | **Community Health Plan of Washington** Aug 2011 – Jun 2012

Software Engineer | **Xperience Communications** Aug 2010 – Aug 2011

// SKILLS

AI / Agentic: LLM applications, voice and conversational agents, multi-agent orchestration, tool use, context engineering, structured outputs, Amazon Bedrock, Codex

Evaluation / Safety: Agent evaluations, goldsets, regression harnesses, guardrails, deterministic verification, tool authorization, execution isolation, provenance, human approval

Engineering: Python, Go, TypeScript, React, Node.js/Express, REST APIs, PostgreSQL, Redis, scikit-learn, Playwright, distributed systems

Cloud / Production: AWS (ECS, Bedrock, IAM, VPC, CloudFront, S3), Terraform, Docker, CI/CD, observability, reliability engineering